BKP: An R Package for Beta Kernel Process Modeling

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2026.07.24

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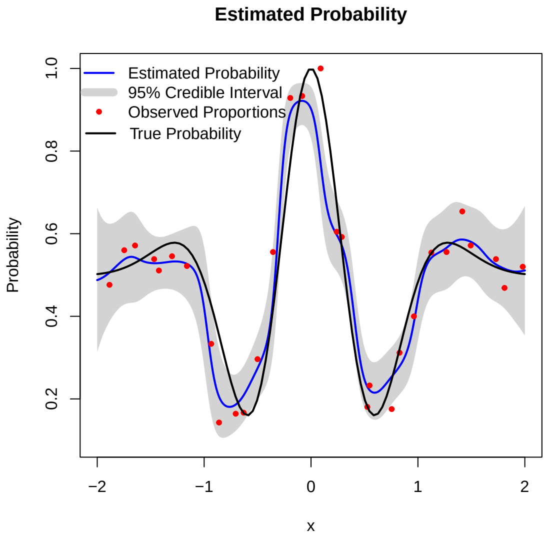
Estimating an Unknown Probability Surface



- **Observation model**

$$y_i \mid \pi(\boldsymbol{x}_i) \sim \mathrm{Binomial}(m_i, \pi(\boldsymbol{x}_i)) \, .$$

- **Objective**
  Estimate the underlying probability surface $\pi(\boldsymbol{x})$ and quantify its pointwise uncertainty.

Existing Approaches and the Remaining Gap

Logistic regression

$$\pi(\boldsymbol{x}) = \text{logit}^{-1}\left(\boldsymbol{x}^\top \boldsymbol{\beta}\right)$$

Fast and interpretable

Limited nonlinear flexibility

Gaussian process classification

$$\pi(\boldsymbol{x}) = \text{logit}^{-1}\{f(\boldsymbol{x})\}, \quad f(\boldsymbol{x}) \sim \text{GP}$$

Flexible probability surfaces

Approximate posterior inference

Motivation

Can we retain nonlinear flexibility while performing direct, closed-form updating on the probability scale?

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Beta Kernel Process: Direct Probability-Scale Updating

For each input location $\boldsymbol{x}$, combine a Beta prior with kernel-weighted binomial evidence.

1. Beta prior

$$\pi(\boldsymbol{x}) \sim \mathrm{Beta}\big\{\alpha_0(\boldsymbol{x}), \beta_0(\boldsymbol{x})\big\}.$$

Prior information on the probability scale

$+$

2. Local likelihood

$$k_i(\boldsymbol{x}) = k(\boldsymbol{x}, \boldsymbol{x}_i; \boldsymbol{\theta}),$$

$$S(\boldsymbol{x}) = \textstyle\sum_{i=1}^{n} k_i(\boldsymbol{x}) y_i,$$

$$F(\boldsymbol{x}) = \textstyle\sum_{i=1}^{n} k_i(\boldsymbol{x})(m_i - y_i).$$

Nearby observations receive greater weight

$\Rightarrow$

3. Beta posterior

$$\pi(\boldsymbol{x}) \mid \mathcal{D} \sim \mathrm{Beta}\big\{\alpha_n(\boldsymbol{x}), \beta_n(\boldsymbol{x})\big\},$$

$$\alpha_n(\boldsymbol{x}) = \alpha_0(\boldsymbol{x}) + S(\boldsymbol{x}),$$

$$\beta_n(\boldsymbol{x}) = \beta_0(\boldsymbol{x}) + F(\boldsymbol{x}).$$

Closed-form posterior inference

Key idea

Local kernel weighting $+$ **Beta–binomial conjugacy**

$\Longrightarrow$ **flexible modeling with closed-form posterior inference**.

Closed-Form Posterior Summaries

Probability estimation

$$\widehat{\pi}(\boldsymbol{x}) = \frac{\alpha_n(\boldsymbol{x})}{\alpha_n(\boldsymbol{x}) + \beta_n(\boldsymbol{x})}$$

Posterior mean of the unknown probability surface

Pointwise uncertainty

$$\operatorname{Var}\{\pi(\boldsymbol{x}) \mid \mathcal{D}\} = \frac{\widehat{\pi}(\boldsymbol{x})\{1 - \widehat{\pi}(\boldsymbol{x})\}}{\alpha_n(\boldsymbol{x}) + \beta_n(\boldsymbol{x}) + 1}$$

$$\mathrm{CI}_{1-\delta}(\boldsymbol{x}) = \left[Q_{\delta/2}(\boldsymbol{x}), Q_{1-\delta/2}(\boldsymbol{x})\right]$$

Credible intervals follow from Beta posterior quantiles

Count prediction

$$Y_{\text{new}} \mid \mathcal{D}, \boldsymbol{x}, m_{\text{new}} \sim \text{Beta-Binomial}\left\{m_{\text{new}}, \alpha_n(\boldsymbol{x}), \beta_n(\boldsymbol{x})\right\}.$$

Closed-form prediction for future binomial counts

Computational implication

Conditional on the kernel hyperparameters, posterior summaries and predictive distributions are available without numerical posterior approximation.

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Summary

- Developed the **first publicly available BKP package** implementing BKP and DKP models
- **Methodological contributions:**
  - Provided a **Bayesian interpretation** of the BKP model
  - Proposed a **flexible framework for constructing priors**, including data-adaptive options
  - Developed a **robust and efficient hyperparameter optimization strategy** using LOOCV and multi-start reparameterization.

Future Development Directions

- **Model extensions**: Extend the BKP framework to multivariate, functional, and spatio-temporal data, as well as mixed-type covariates
- **Alternative likelihoods**: Explore likelihoods beyond the binomial family, e.g., negative binomial for over-dispersed counts, geometric for waiting-time modeling
- **Community contributions**: We invite developers to contribute via GitHub pull requests `https://github.com/Jiangyan-Zhao/BKP/pulls`

Resources

- https://cran.r-project.org/web/packages/BKP/index.html
- https://github.com/Jiangyan-Zhao/BKP
- https://arxiv.org/abs/2508.10447


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THANK YOU